



WRD Series

Refrigerated Compressed Air Dryers

Compressed air leaves the compressor saturated with water. WRD dryers chill it to a +3°C pressure dew point so the moisture condenses and drains out before it reaches your machines, valves and finished product.

Capacity

10 to 2000 CFM

Pressure dew point

+3 °C

Inlet air pressure

3 to 16 bar g

Inlet air temperature

10 to 60 °C

Win Equipments

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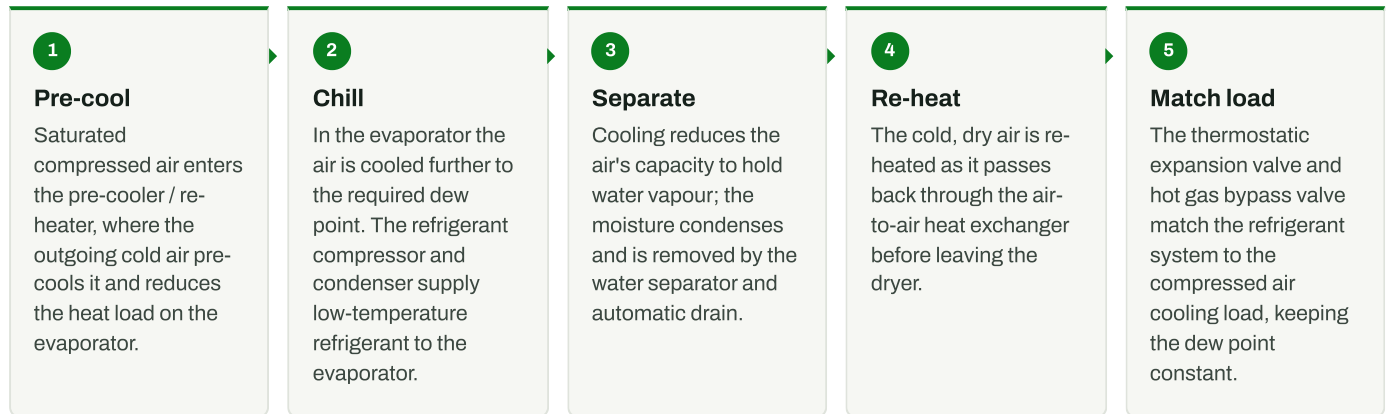
Full model range,
photos and
sizing tools



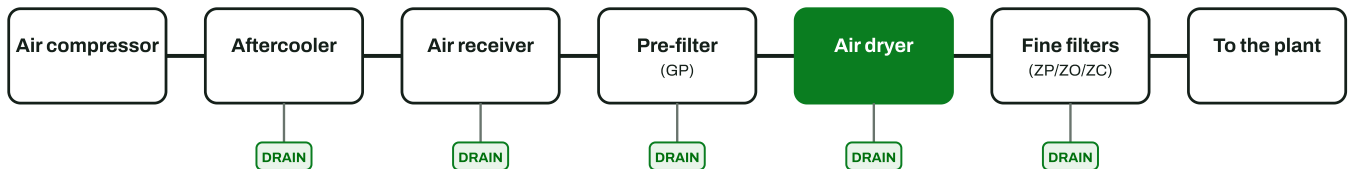
WORKING PRINCIPLE

How it works

Air contains dust, moisture and traces of oil. Untreated wet compressed air causes equipment failures and contaminates the production process, so it must be dried before use.



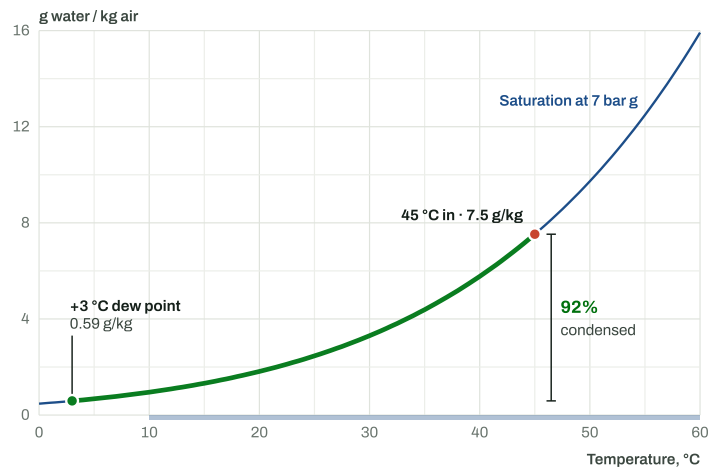
TYPICAL INSTALLATION



Automatic drain valves discharge condensate from the aftercooler, receiver, filters and dryer.

WHY DRY THE AIR

Water held by compressed air at 7 bar g



Saturated air at 45 °C holds about 7.5 g of water per kg; at a +3 °C pressure dew point only about 0.6 g/kg, so about 92% must condense and drain. Calculated values.

Features

Co-axial heat exchanger

- Acts as pre-cooler and evaporator
- Copper tubes in a coiled tube-in-tube arrangement
- Fully encapsulated in PUF insulation

Temperature controller

- Programmed micro-controller
- Indicates dew point, inlet and condensing temperature with alarms
- On-delay timer protects the compressor

Refrigeration compressor

- Hermetically sealed reciprocating or scroll compressor
- Suitable for eco-friendly gases
- Low noise and low power consumption

Hot gas bypass valve

- Fitted between compressor discharge and evaporator
- Prevents evaporator temperature dropping below 5 °C
- Manual or automatic

Thermostatic expansion valve

- Selected for the cooling load and operating condition
- Holds a constant dew point at varying load
- Responds to suction pressure and temperature

Condenser

- Air-cooled: aluminium plate-finned cross-flow exchanger
- Water-cooled option on larger models
- Spigot construction lowers pressure drop and compressor power

WRD Series models

Rating conditions

Flow rating ISO 7183, FAD at 20 °C, 1 bar	Max. working pressure 16 bar g	Pressure drop ≤ 0.2 bar g	Supply 180–260 V 1 Ph · 380–420 V 3 Ph
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Model	Capacity CFM	Capacity m³/hr	Refrigeration compressor kW	Electrical V / Ph / Hz	Dimensions L × W × H mm	Condenser	End connections BSP / NB
WRD 20 S	20	34	0.13	230 / 1 / 50	550 × 400 × 600	Air	½"
WRD 40 S	40	68	0.21	230 / 1 / 50	550 × 400 × 600	Air	½"
WRD 50 S	50	85	0.28	230 / 1 / 50	540 × 500 × 600	Air	¾"
WRD 60 S	60	102	0.32	230 / 1 / 50	540 × 500 × 800	Air	¾"
WRD 80 S	80	136	0.40	230 / 1 / 50	540 × 500 × 800	Air	1"
WRD 100 S	100	170	0.67	230 / 1 / 50	700 × 500 × 800	Air	1"
WRD 150 S	150	255	0.85	230 / 1 / 50	700 × 500 × 1000	Air	1½"
WRD 200 S	200	340	1.2	230 / 1 / 50	700 × 750 × 1000	Air	1½"
WRD 250 S	250	425	1.35	230 / 1 / 50	900 × 750 × 1000	Air	2"
WRD 300 T	300	510	1.7	440 / 3 / 50	900 × 750 × 1200	Air	2"
WRD 400 T	400	680	2.0	440 / 3 / 50	950 × 1700 × 1200	Air	2½"
WRD 500 T	500	850	2.3	440 / 3 / 50	1450 × 1700 × 1250	Air	2½"
WRD 600 T	600	1020	2.72	440 / 3 / 50	1450 × 1700 × 2100	Air / Water	2½"
WRD 750 T	750	1275	3.3	440 / 3 / 50	1450 × 1700 × 2100	Air / Water	3" NB flange
WRD 1000 T	1000	1700	4.5	440 / 3 / 50	1450 × 1700 × 2100	Air / Water	4" NB flange
WRD 1500 T	1500	2550	6.2	440 / 3 / 50	1450 × 1700 × 2100	Air / Water	5" NB flange
WRD 2000 T	2000	3400	7.8	440 / 3 / 50	1450 × 1700 × 2100	Air / Water	6" NB flange

- › Model code: WRD + capacity in CFM + S (single phase) or T (three phase). Working pressure 16 bar g max.
- › Flow capacities per ISO 7183: air suction of FAD at 20 °C, 1 bar.
- › Rated power is the maximum power consumed at rated conditions.
- › Voltage range 180–260 V for 1 Ph and 380–420 V for 3 Ph.

Sizing for your conditions

$$\text{Dryer nominal capacity} = \text{Compressor actual capacity} \div (\text{C1} \times \text{C2} \times \text{C3} \times \text{C4})$$

C1 · Inlet temperature

°C	30	35	40	45	50	55	60
Factor	1.2	1.15	1.05	1.0	0.85	0.8	0.7

C2 · Ambient temperature

°C	25	30	35	40	45	50
Factor	1.2	1.15	1.0	0.91	0.87	0.78

C3 · Inlet pressure

bar g	4	5	6	7	8	9	10.5	11	12	13	14	15
Factor	0.75	0.85	0.93	1.0	1.06	1.11	1.15	1.18	1.2	1.22	1.23	1.25

C4 · Dew point

°C	3	5	7	8	10
Factor	1.0	1.09	1.15	1.18	1.3

APPLICATIONS

Automobile

Cement

Chemical

CNC machine shops

Electronics

Food processing

Foundry

Glass

Leather

Textile

Paper mills

Rice mills

Power plants

Printing

QUESTIONS

What dew point does a WRD dryer deliver?

WRD dryers deliver a +3 °C pressure dew point and hold it at varying load using a thermostatic expansion valve and hot gas bypass valve.

How do I size a refrigerated dryer for my compressor?

Divide your compressor's actual capacity by the product of the four correction factors for inlet temperature, ambient temperature, inlet pressure and dew point. Then choose the WRD model at or above that nominal capacity. The sizing calculator on winequipments.com does the calculation for you.

Which models run on single-phase power?

Models with an S suffix (WRD 20 S to WRD 250 S) run on 230 V single phase. T models from WRD 300 T upward run on three-phase supply.

Is a water-cooled condenser available?

Yes. WRD 600 T and larger are offered with air- or water-cooled condensers.

Talk to an engineer

Send your compressor capacity, heat load or water flow, operating pressure, temperatures and site conditions. We reply with a recommended model and quotation.

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ISO 9001:2015 certified. Specifications are as per our catalogue and are subject to change without notice; confirm the selection with our engineers before ordering. Issued October 2026.